

AI in Data Analysis: Empowering PruBSN with Copilot in Excel

From Raw Data to Insights in Minutes



Dr. Dwi Pebrianti

Module 3

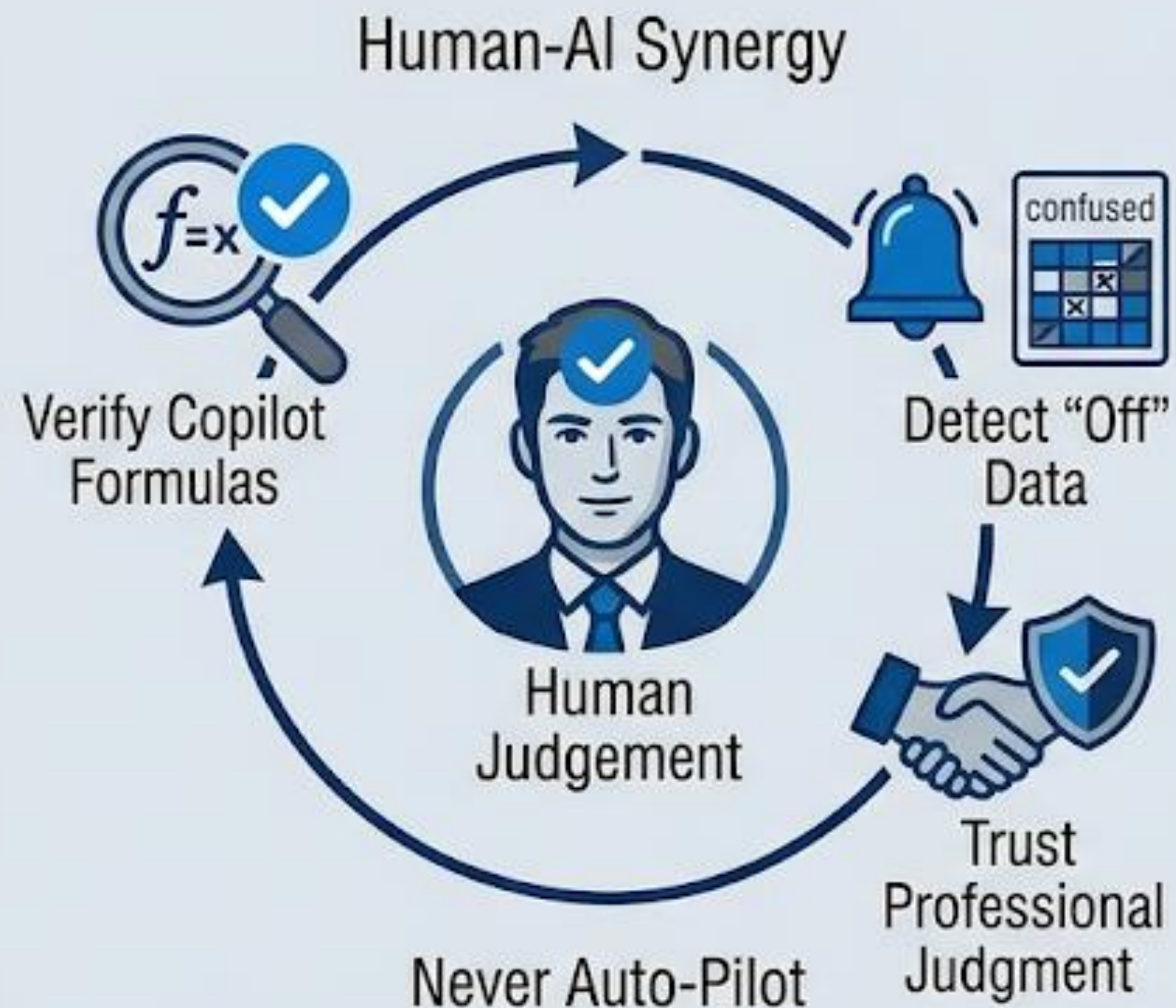
Copilot is your analytical assistant, not a replacement for your expertise

Data Analysis Philosophy: The Copilot Paradigm Shift

The “80/20” Rule: The Historical Shift



Human-in-the-Loop: Trust and Validate



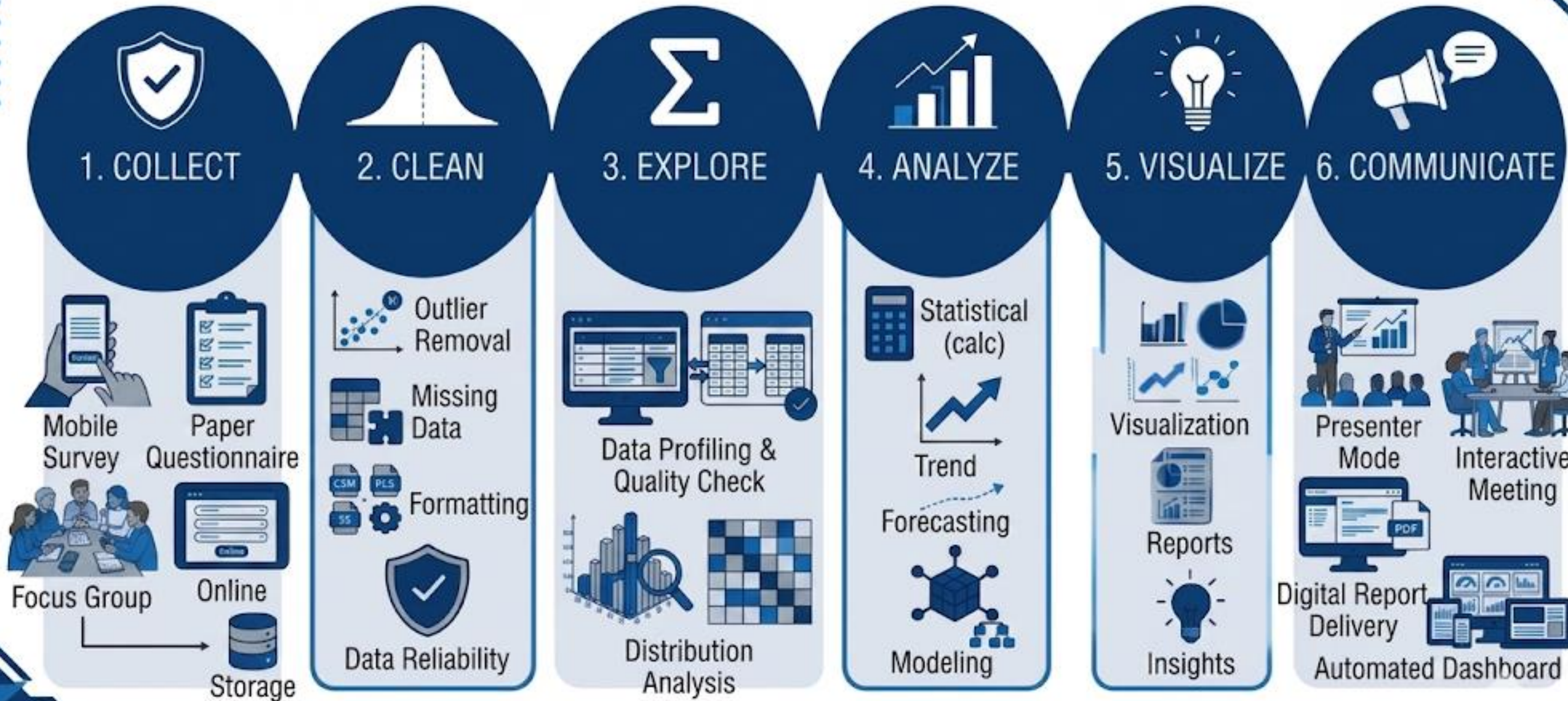
Copilot empowers your expertise. You are the critical, human element for strategic insights.

DATA CYCLE: AN INTEGRATED GUIDE

Foundation

Transformation

Impact





HANDS ON

Step 1: Data Collection



Link to sample file

<https://github.com/aimighazali/TrainingPruBSN>

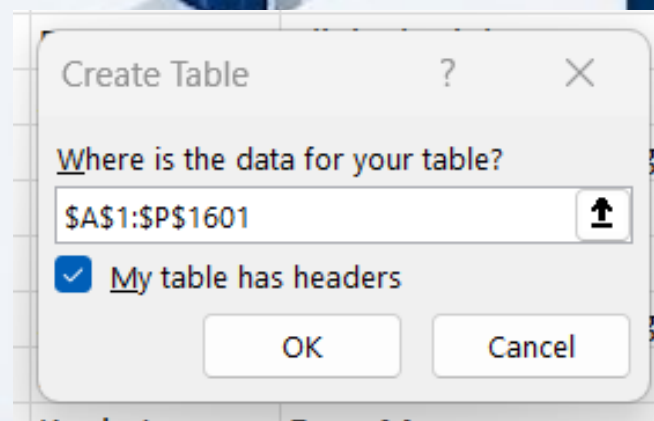
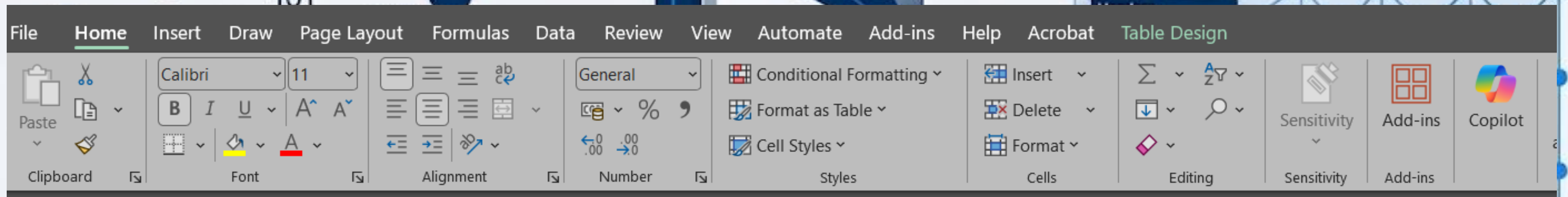
 Facttable.xlsx	Add files via upload	last month
 OrderCalendar.txt	Add files via upload	last month
 PRUBSN_11thMay_BusinessData.xlsx	Add files via upload	2 months ago
 PRUBSN_Tier 1_Business Data.xlsx	Add files via upload	1 minute ago
README		



Ctrl + T is the essential ignition key for your AI assistant.

Unreadable by AI

Headers, data types, and relationships unlocked.



Why? Copilot needs a structured Table to understand the context of your data. It cannot 'read' a loose range of cells.



HANDS ON

Step 2: Data Cleaning



Data Cleaning Essentials

What is Data Cleaning?

Detecting and correcting inaccurate, incomplete, or inconsistent data in a dataset

Why Clean Data?

To improve data quality — accurate, reliable, and ready for analysis

Key Data Cleaning Steps

- Remove errors, blanks, or duplicates
- Replace incorrect values
- Change or fix data types
- Standardize text formats and column names
- Handle missing or null data



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Common errors



movie_title	num_critic_for_reviews	duration	DIRECTOR_facebook_likes	actor_3_facebook_likes	ACTOR_1_facebook_likes	gross
Avatar?	723	178	10	855	1000	760505847
Pirates of the Caribbean: At World's End?	302		563	1000	40000	309404152
Spectre?	602	148	20	161	11000	200074175
The Dark Knight Rises?	813		22000	23000	27000	448130642
John Carter?	462	132	"475"	530	640	73058679
Spider-Man 3?	392	156	23	4000	24000	336530303
Tangled?	324		15	284	799	200807262
Avengers: Age of Ultron?	635	141	10	19000	26000	458991599
Avengers: Age of Ultron?	635	141	10	19000	26000	458991599
Harry Potter and the Half-Blood Prince?	375	153	282	10000	25000	301956980
Batman v Superman: Dawn of Justice?	673	183		2000	15000	330249062
Superman Returns?	434	169		903	18000	200069408
Quantum of Solace?	403	106	395	393	451	168368427

Inconsistent headers

Noisy data

Incorrect formatting

Duplication

Missing data





Ask co-pilot:

For data cleaning, analyze the data:
What is wrong with it?
Are there any errors?

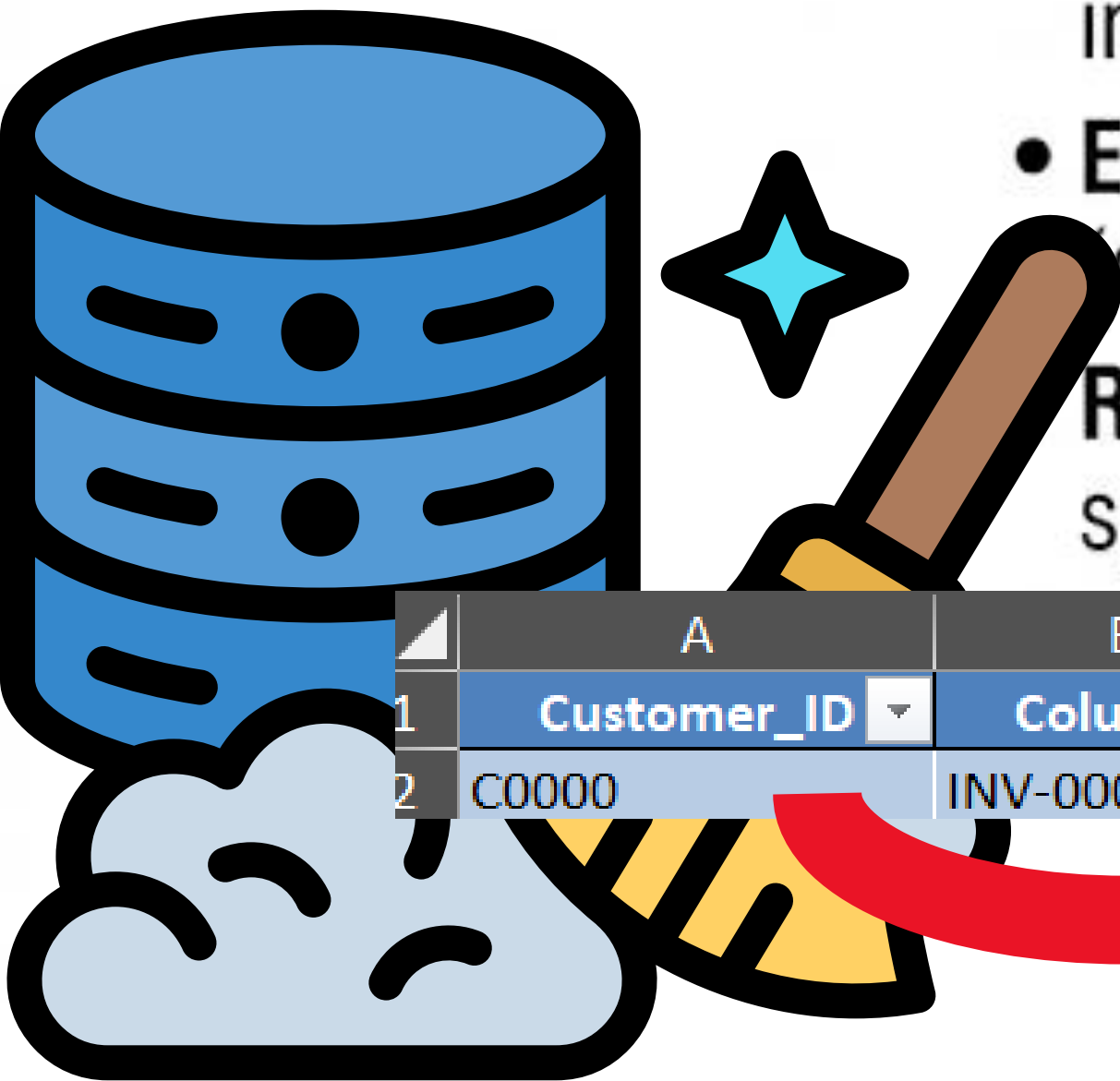


LET'S FIX THE ERROR:

Challenge 1 - Column Renaming.

Why is this Important? (Benefits)

- **IMPROVES CLARITY AND UNDERSTANDING:** Use meaningful names like CustomerName or SalesAmount instead of vague ones like Col1 or Value.
- **ENSURE CONSISTENCY:** Standardize data formats (e.g., STD_ID) across all tables and databases.
- **REFLECTS TRUE MEANING:** Change generic terms to specific ones, like Vague name... to Total_Sales.



	A	B	C	D	E	F
1	Customer_ID ▾	Column1 ▾	Customer_Name ▾	Region ▾	Product ▾	Quantity ▾
2	C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7

LET'S FIX THE ERROR:

Challenge 1 - Column Renaming.

Co-pilot: Find column headers with generic labels and suggest appropriate labels for them.

Inappropriate Column Name	Suggested Appropriate Name
Col1	Customer_ID
A	Order_Date
Data_2023	Sales_Amount

12 cols
to review



LET'S FIX THE ERROR:

Challenge 2 - Duplicate Data.

Identical or repeated records Identical or repeated records appearing more than once in the dataset.

Causes of Duplicate Data:

Main Causes	Implications (Summary)
(i) Repeated data imports or merges (ii) Manual entry errors (iii) Multiple sources collecting the same information (iv) System glitches or unsynchronized databases	• Identical records appear more than once. • Leading to distorted results in analysis and reporting. • Causes discredit and poor decision making. • Increases storage costs.



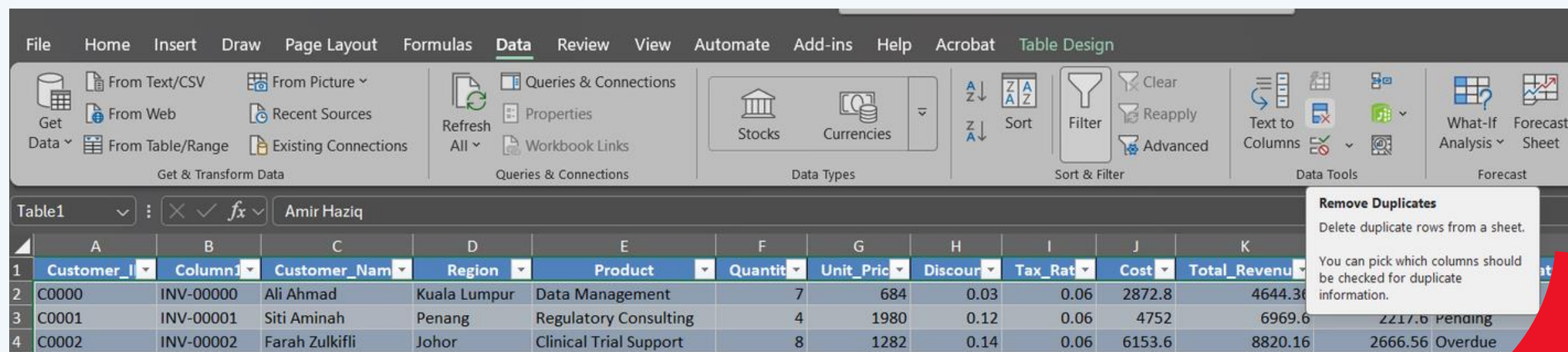
Prompt: Find duplicate data

✓ How to handle this in Excel (quick steps)

If you want to remove duplicates directly in Excel:

1. Select the entire table
2. Go to Data → Remove Duplicates
3. Select all columns
4. Click OK

Excel will remove exactly one duplicate row
(the C0100 / INV-00100 entry).



	A	B	C	D	E	F	G	H	I	J	K
1	Customer_ID	Column1	Customer_Nam	Region	Product	Quantit	Unit_Pric	Discour	Tax_Rat	Cost	Total_Revenu
2	C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7	684	0.03	0.06	2872.8	4644.36
3	C0001	INV-00001	Siti Aminah	Penang	Regulatory Consulting	4	1980	0.12	0.06	4752	6969.6
4	C0002	INV-00002	Farah Zulkifli	Johor	Clinical Trial Support	8	1282	0.14	0.06	6153.6	8820.16



Data Cycle Guide: Addressing Missing Values (Clean)

Challenge 3: Missing Values

Data gaps where information is blank, null, or unreadable
(e.g., N/A, empty cells)

2. CLEAN

Address structure
& errors

REASON ANALYSIS

(Causes of Data Gaps)

- **Human error**
Incorrect data entry, typos, misreading info
- **System or sensor failure**
Sensor malfunctions, power loss, database connectivity issues
- **Conditional or intentional skipping**
Skip logic in surveys, deliberate non-response
- **Privacy or anonymity constraints**
Data masking, removal for compliance & privacy

REMEDIATION STRATEGIES

(Techniques for Imputation & Handling)

- **Remove rows** – if few or irrelevant
- **Replace values** – with mean, median, or "Unknown"
- **Fill up / down** – for sequential data
- **Flag missing** – keep track for later analysis

Why? Copilot needs structure. Imputation or flagging provides **necessary structure.**

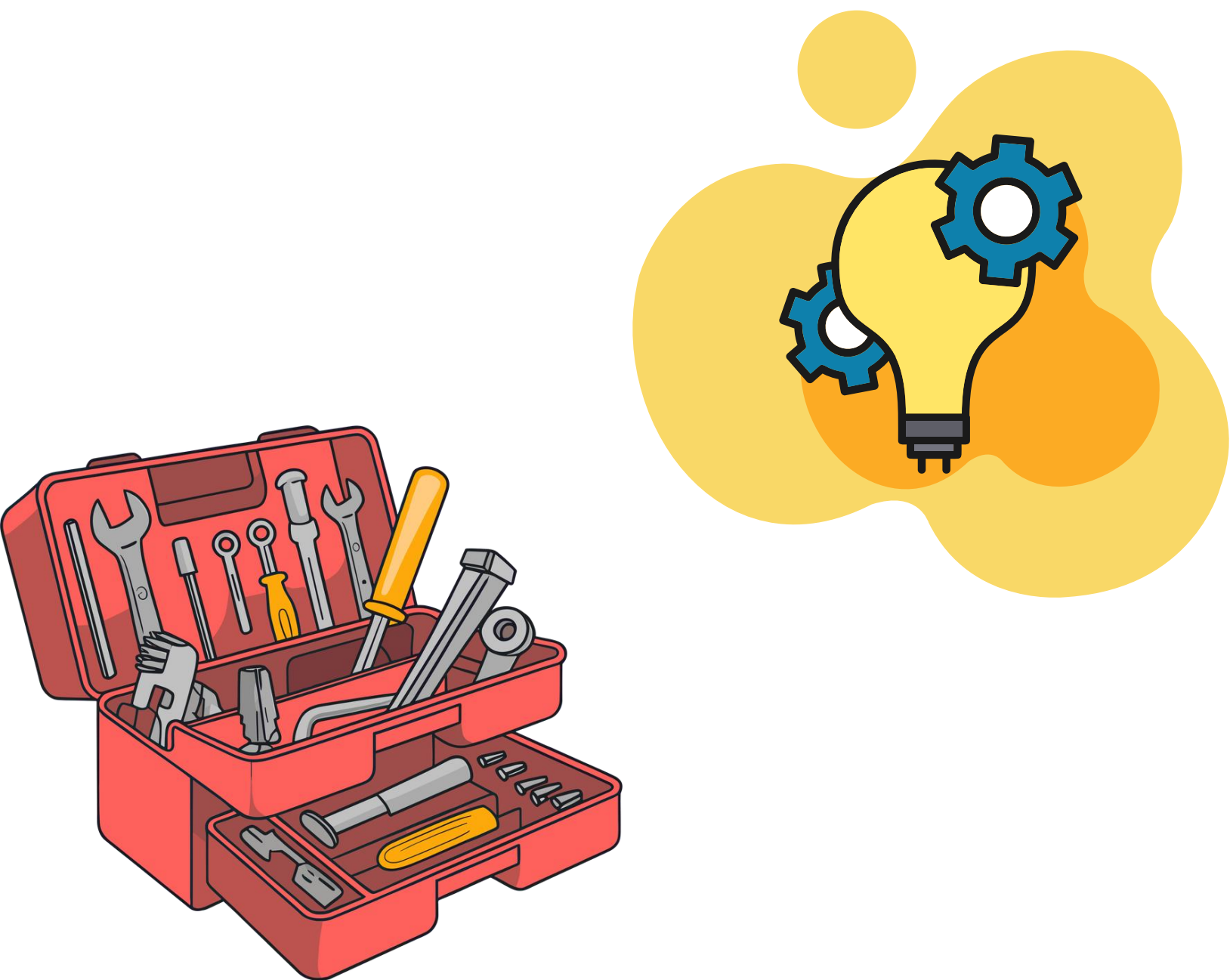


This is *your* expertise!

I have analyzed your data file and can confirm that **Row 402** (which corresponds to Customer ID **C0400**, Invoice **INV-00400**) is indeed missing a value in the **Discount** column.

Verification details for Row 402:

- **Product:** Regulatory Consulting
- **Quantity:** 1
- **Unit Price:** 983
- **Total Revenue:** 707.76
- **Discount:** *Missing (NaN)*



Suggested Best Way to Fill the Value

In your dataset, the values follow a specific mathematical relationship. Instead of using a simple average (mean) or median, the most accurate way is to **reverse-calculate** the discount using the existing financial data in that row.

Based on the other rows in your file, the formula used is:

$$\text{Total Revenue} = (\text{Quantity} \times \text{Unit Price}) \times (1 - \text{Discount})$$

The Calculation:

1. **Gross Amount:** $1 \times 983 = 983$
2. **Revenue Ratio:** $707.76 / 983 = 0.72$
3. **Discount Value:** $1 - 0.72 = 0.28$

Recommended Value: You should enter **0.28** (or 28%) into the Discount cell for Row 402. This perfectly aligns with the Total Revenue and Net Income already recorded for that entry.

How to apply this in Excel/CSV:

1. Locate cell **H402** (assuming Column H is Discount).
2. Type `0.28`.
3. If you want to automate this for any future missing values, you can use this Excel formula in the Discount column:

```
=1 - (Total_Revenue / (Quantity * Unit_Price))
```



HANDS ON

Step 3: Data Exploration

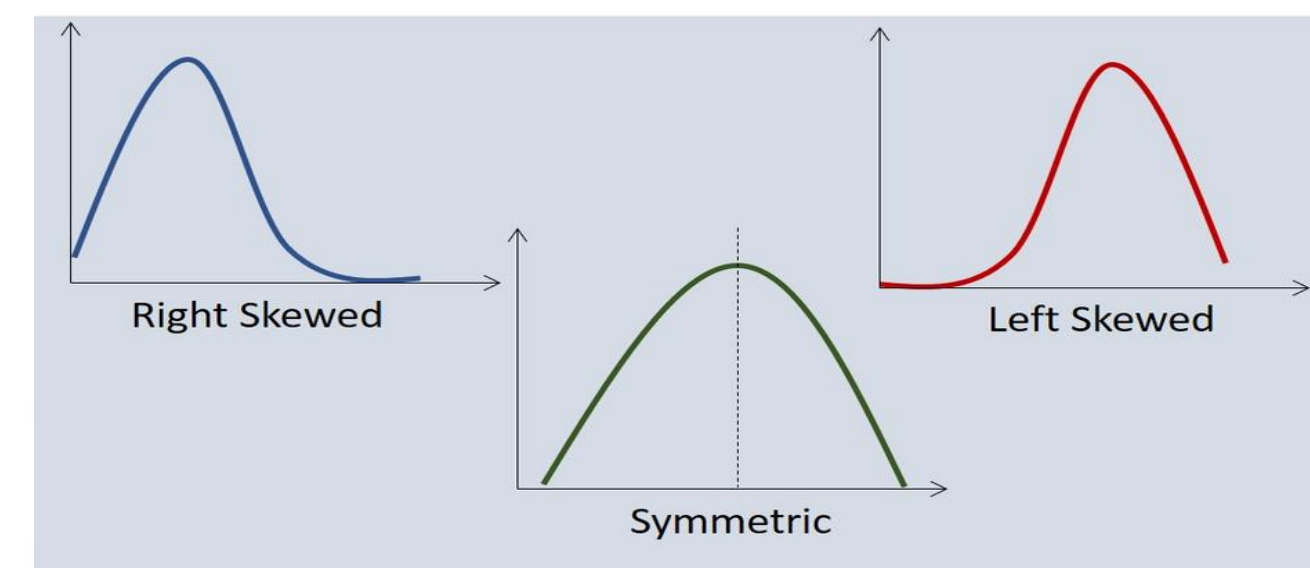
Please perform a distribution analysis.

Share the outputs.

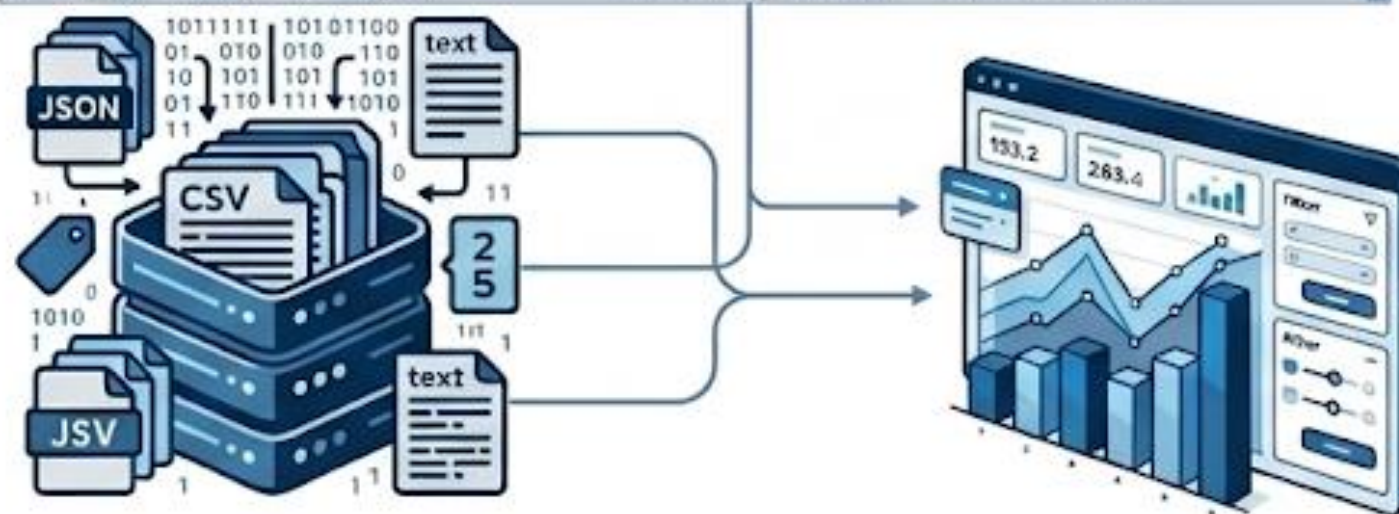




Outcomes:



Field Type	Distribution Shape
Regions	Moderately imbalanced
Products	Balanced (3 categories)
Payment Status	Balanced (rule-based)
Quantity	Right-skewed
Prices & Revenue	Heavily right-skewed
Tax Rate	Narrow, stable
Delivery Days	Compact, uniform



Overall Data Quality Assessment

✓ Strengths

- Rich transactional detail
- High completeness
- Logical relationships preserved
- Suitable for customer, revenue, and operational analysis

✗ Weaknesses

- Numeric formatting issues
- Scientific notation artifacts
- Outliers without flags
- Minor text inconsistencies

Recommended Next Steps

e.g., Distribution analysis

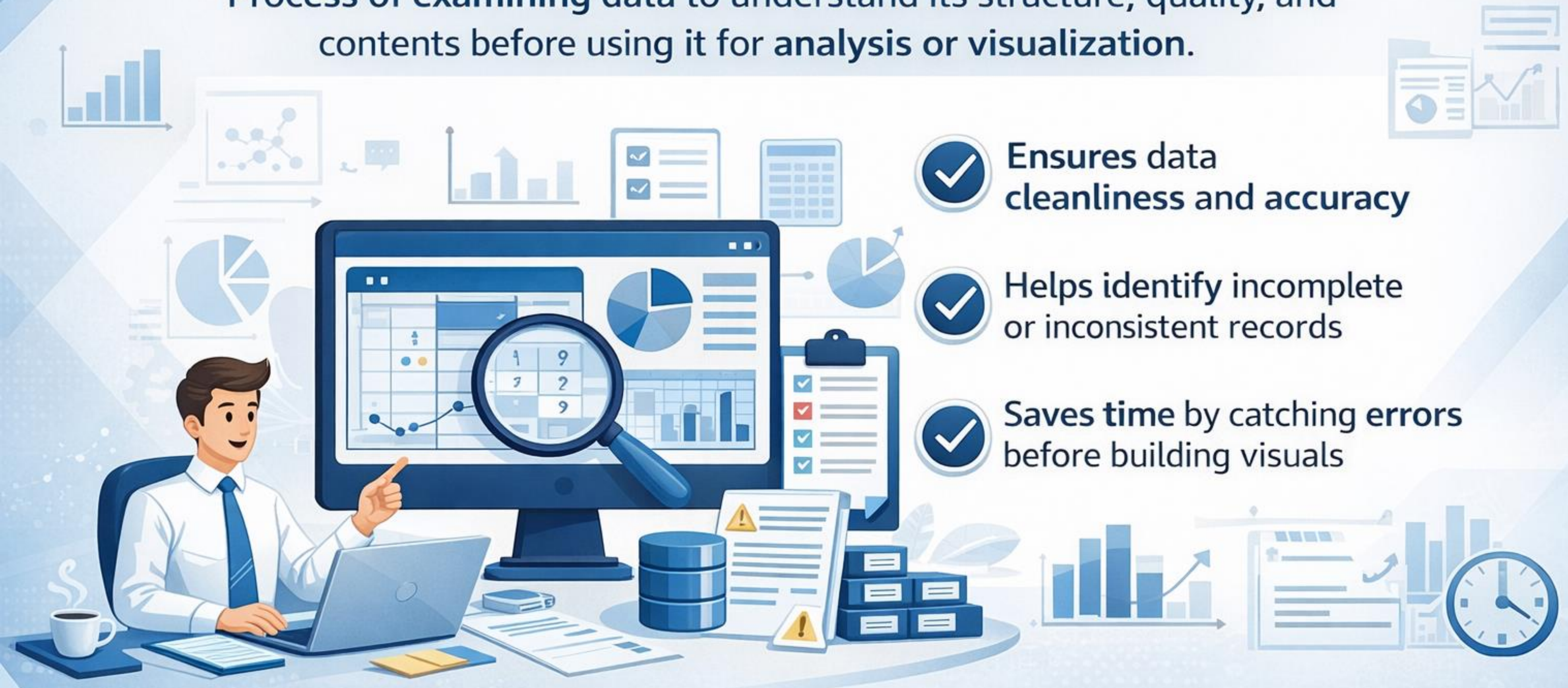
- **Normalize numeric formats:** Convert scientific notation to standard decimals, round monetary fields consistently.
- **Validate outliers:** Apply IQR/Z-score checks on Quantity and Revenue, flag extreme invoices for review.
- **Clean categorical text:** Standardize capitalization, trim and correct corrupted strings.
- **Recompute financials:** Recalculate Cost, Tax, Revenue, Net_Income from base fields, compare against stored values.
- **Prepare for analytics:** Create derived metrics (ARPU, overdue rate, avg delivery time), segment customers by region/service.



Why Data Exploration Matters

Process of examining data to understand its structure, quality, and contents before using it for analysis or visualization.

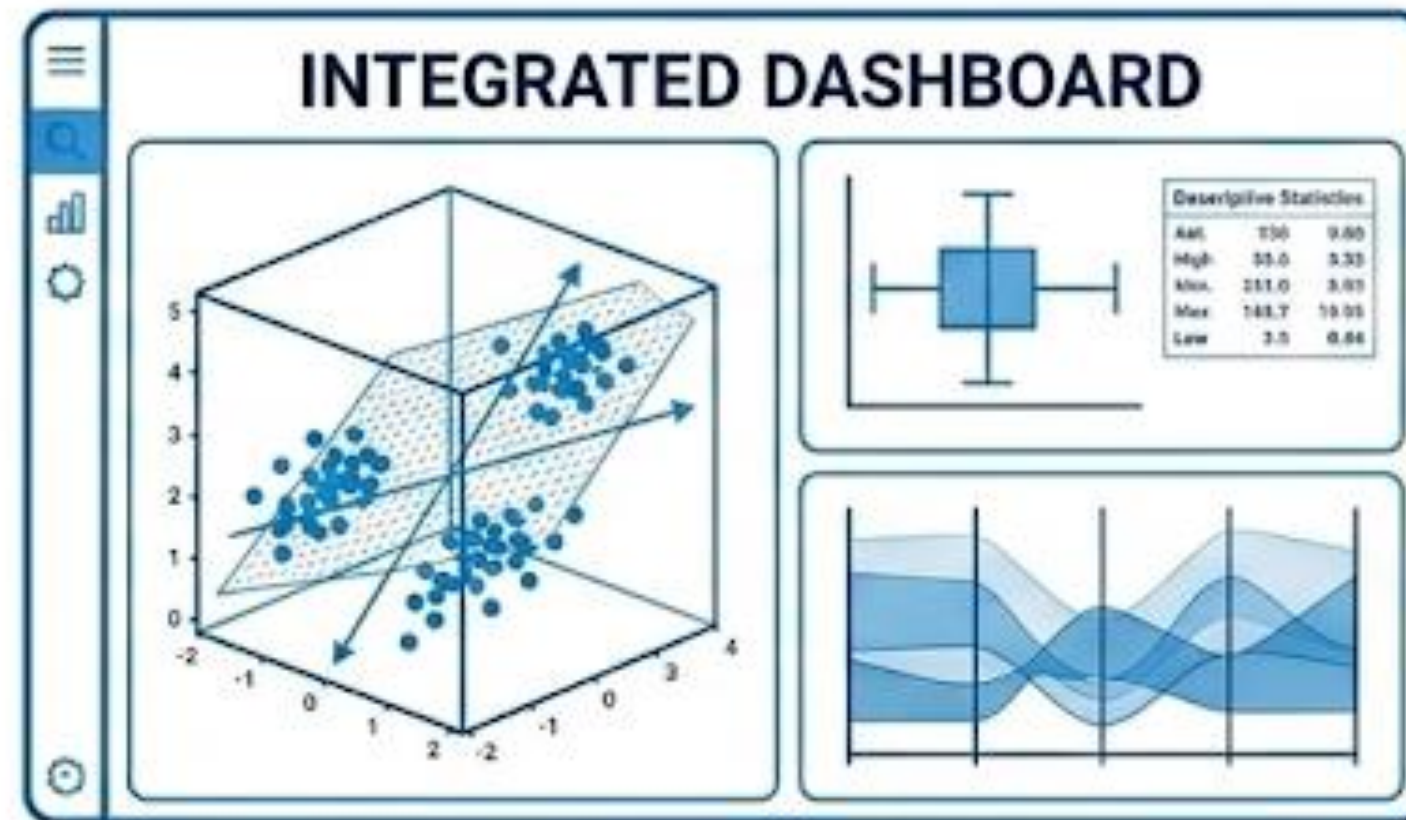
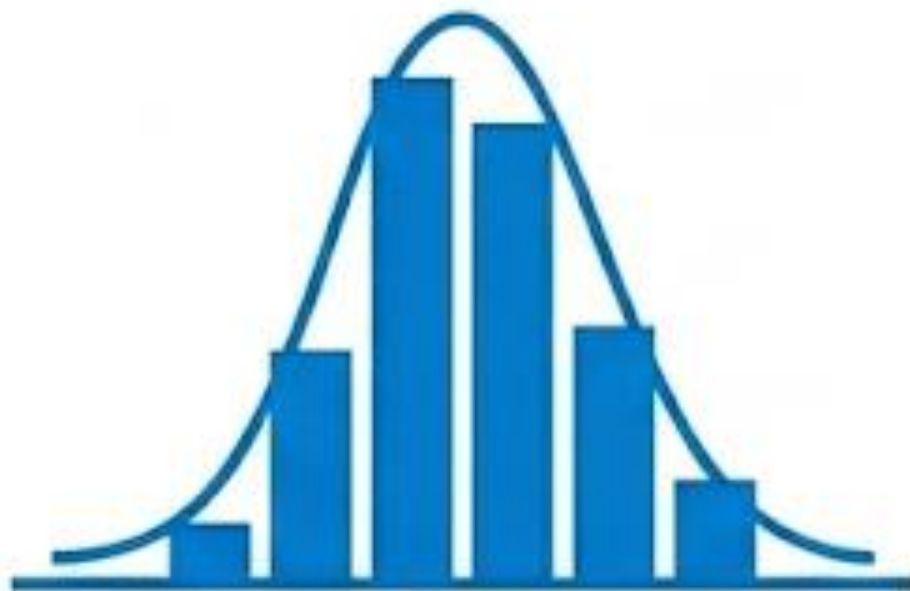
- ✓ Ensures data cleanliness and accuracy
- ✓ Helps identify incomplete or inconsistent records
- ✓ Saves time by catching errors before building visuals





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Step 4 & 5: Data Analysis & Visualization (Focus: Applied Statistics & Advanced Visuals)





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Statistical Model & Forecasting

[Task 1] Revenue Distribution Analysis by Product Type

Analyze how Total_Revenue differs across:

- ProductData Management
- Regulatory Consulting
- Clinical Trial Support

Methods

Descriptive statistics:

- Mean, median, standard deviation of Total_Revenue

Visualization:

- Boxplots or histograms

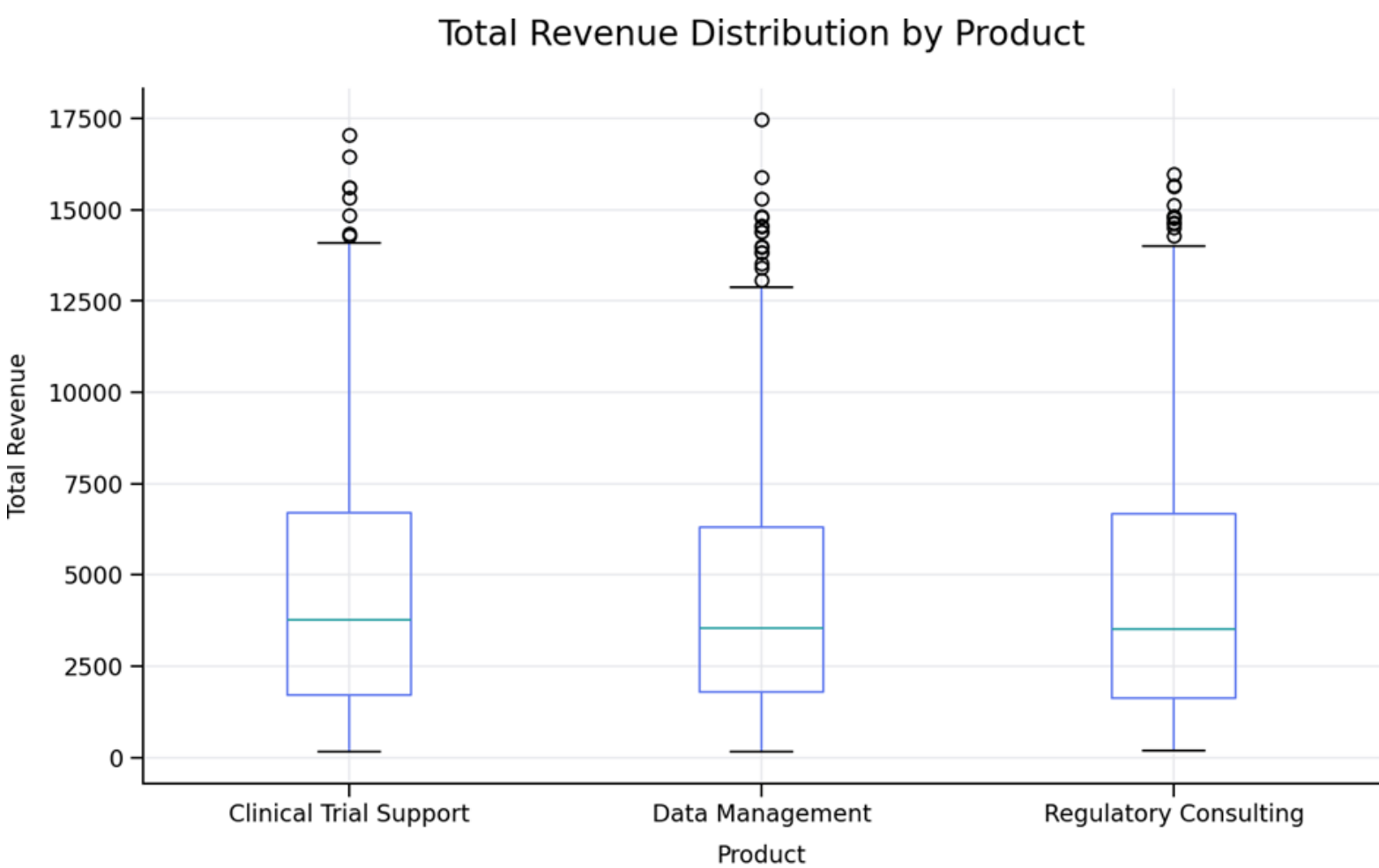


Task 1

Product	Mean	Median	Standard Deviation
Clinical Trial Support	4,653.96	3,774.19	3,504.52
Data Management	4,575.54	3,544.72	3,505.44
Regulatory Consulting	4,560.62	3,522.26	3,521.73

🔍 Interpretation

- Clinical Trial Support has the highest mean and median revenue, indicating slightly stronger revenue performance.
- Median values are consistently lower than means, suggesting right-skewed distributions (presence of high-value outliers).
- Standard deviation is high and similar across all products, indicating:
 - Large variation in invoice sizes
 - Comparable revenue volatility across services



🔍 Boxplot Insights

- All three products show:
 - Wide interquartile ranges
 - Multiple high-value outliers
- Clinical Trial Support exhibits:
 - Slightly higher upper quartile
 - More extreme upper outliers
- Revenue distributions across products are similar in shape, indicating comparable pricing structures with different deal sizes.



HANDS ON



Statistical Model & Forecasting

[Task 2] Forecastable Trend: Future Total Revenue Trend

Trend to forecast:

Total Revenue over invoice sequence or time proxy
(If no explicit date column exists, invoice number order can be used as a proxy for time.)

Forecasting approach

Aggregate:

- Total revenue per invoice batch or period

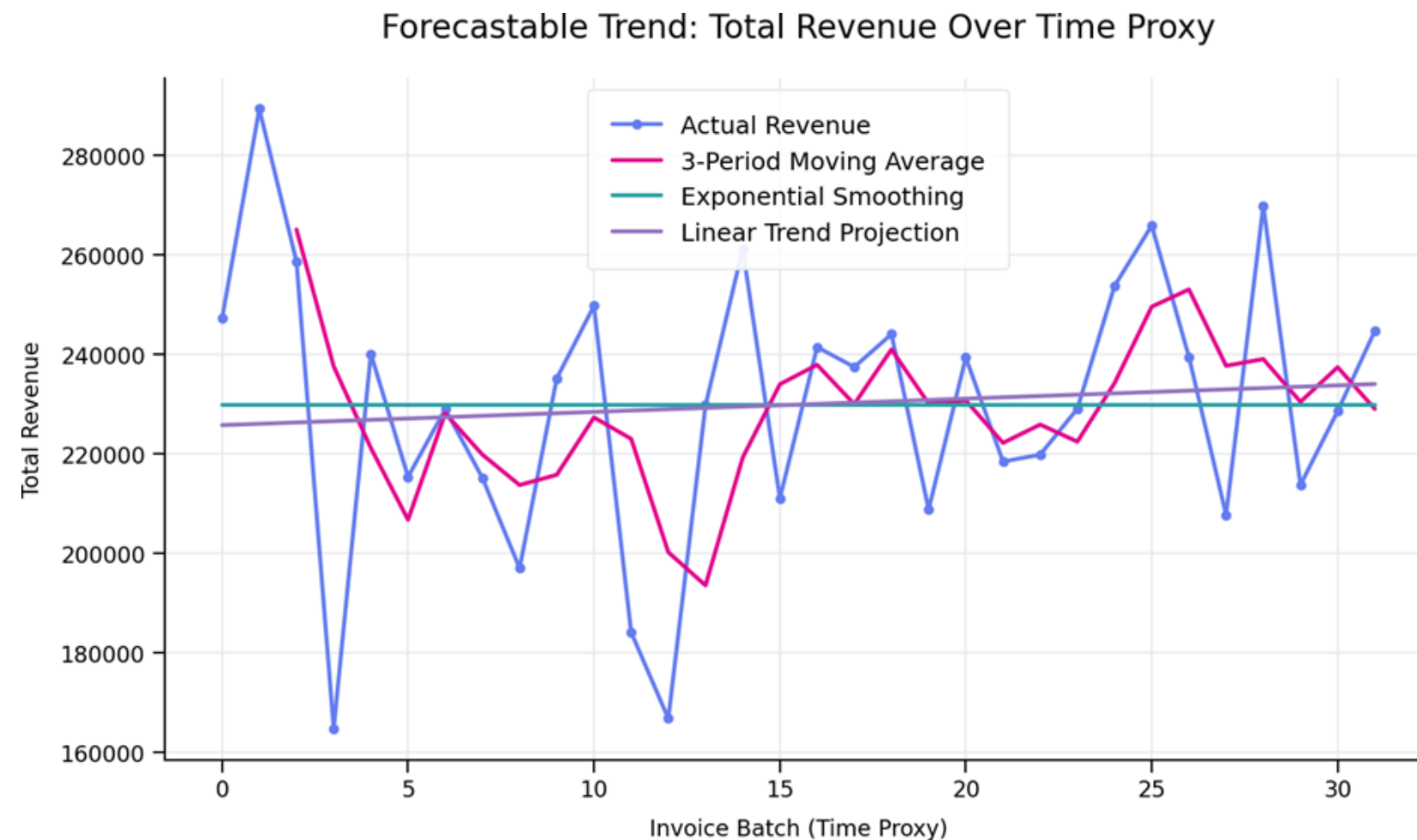
Apply time-series methods:

- Moving average
- Exponential smoothing
- Linear trend projection

Visualization:

- Actual aggregated revenue
- Moving average
- Exponential smoothing
- Linear trend projection

Task 2



✓ Interpretation of Results

🔍 Key Observations

Revenue is generally stable with mild growth
Outliers exist, but aggregation reduces their impact
No evidence of sustained revenue decline
Linear trend suggests incremental long-term growth

🔮 What Can Be Forecasted

Near-term revenue using exponential smoothing
Medium-term revenue trajectory using linear trend
Expected revenue per upcoming invoice batches

✓ Conclusion (Sample)

Using invoice sequence as a time proxy, total revenue exhibits a stable pattern with gradual positive growth. Moving average and exponential smoothing models indicate short-term stability, while linear trend projection suggests modest long-term revenue expansion. These findings support the use of trend-based forecasting for budgeting and sales planning in the absence of explicit temporal data.

 New chat

 Search

 Library

 Create

 Teach

Agents

 Excel

 New agent

... All agents

Popular in your org



PowerPoint (Agent)

Microsoft Corporation

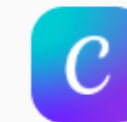
Create a PowerPoint presentation using just your words



Excel (Agent)

Microsoft Corporation

Create an Excel workbook using just your words



Canva

Canva Pty Ltd

Canva is the easy to use, collaborative design platform where teams create together!



Asana

Asana

Turn conversations into action items in Asana to keep work on track



Adobe Express

ADOBE INC.

Create high-quality designs in seconds using text prompts.



Word (Agent)

Microsoft Corporation

Create a Word document using just your words

Upload to Cloud

My files > Documents > Copilot > Created

Make your data work for you, not the other way around.

~~Manual-labor more.~~
~~Analyze less.~~
Analyze more.
Manual-labor less.

AI for Productivity: Automation Using Microsoft Teams Workflows

Work smarter and eliminate
repetitive manual tasks.



Trainer: Assoc. Prof. Ts. Dr. Hasan Firdaus Bin Mohd Zaki
Centre for Unmanned Technologies, International Islamic University Malaysia

Core Objectives for Today's Session



Understand Microsoft Teams Workflows

Discover the no-code engine powered by Power Automate.



Automate Repetitive Tasks

Let Microsoft handle manual follow-ups, scheduling, and notifications.



Create Your First Workflow

Build and test a working automation live in this session.



Apply to Real Scenarios

Identify and deploy automations specific to your role at Prudential Takaful.

Why Automation is a Competitive Advantage



The Core Logic: Trigger, Action, Result



Built natively in Teams. Powered by Microsoft Power Automate. 100% No-code / Low-code.

Locating the Workflows App Inside Teams

1

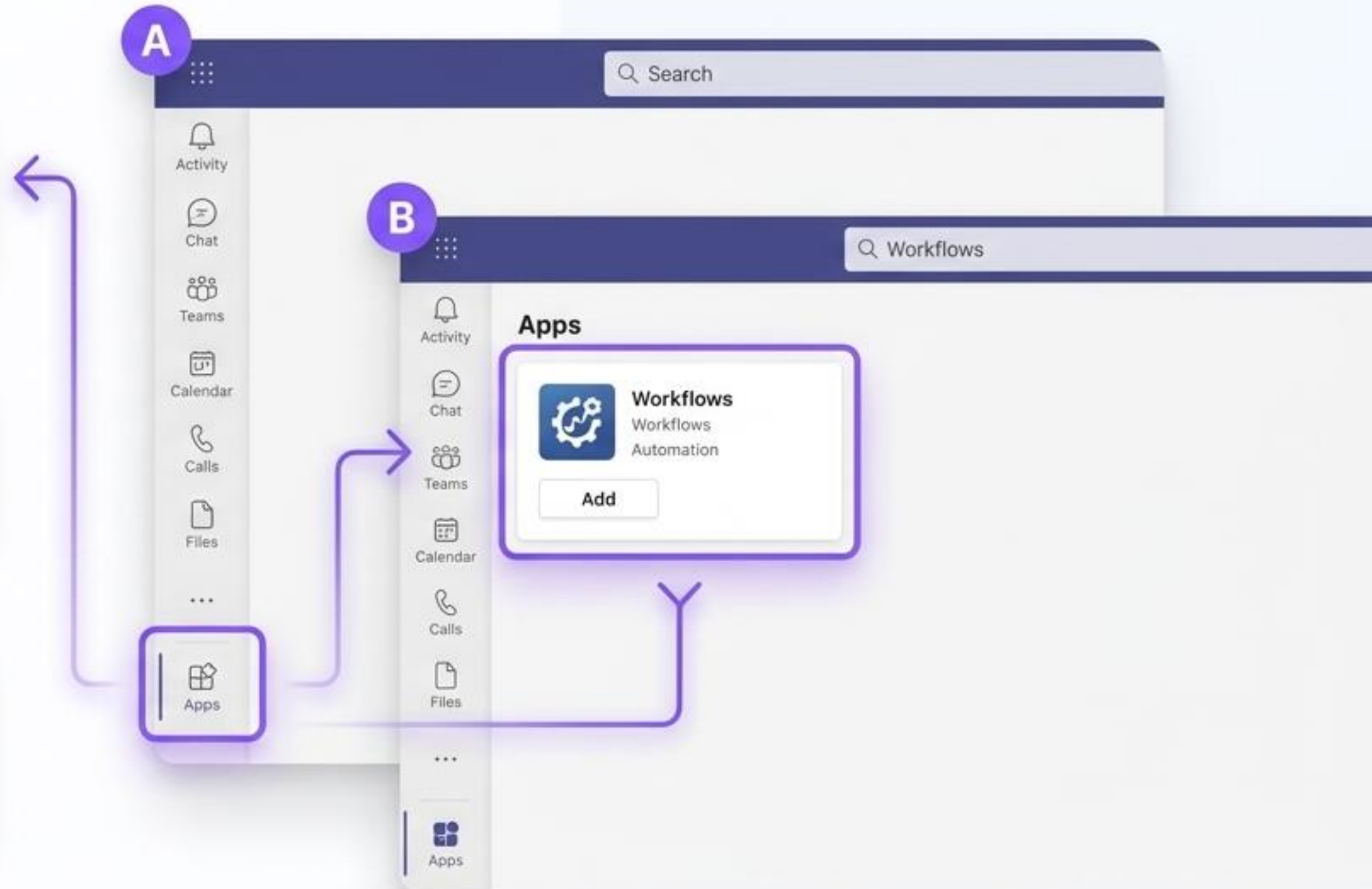
Open Microsoft Teams and click 'Apps' (bottom left corner of the sidebar).

2

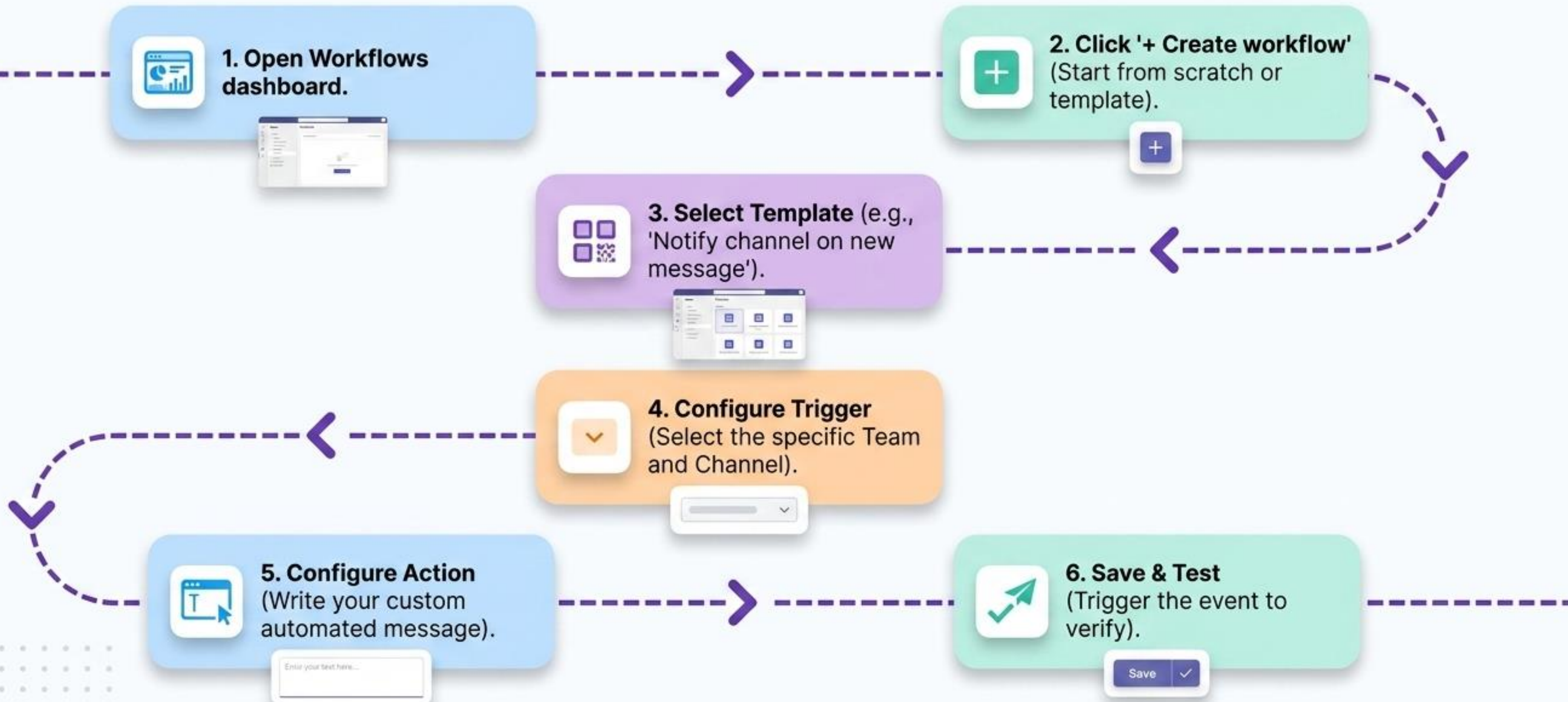
Search for 'Workflows' in the top app search bar.

3

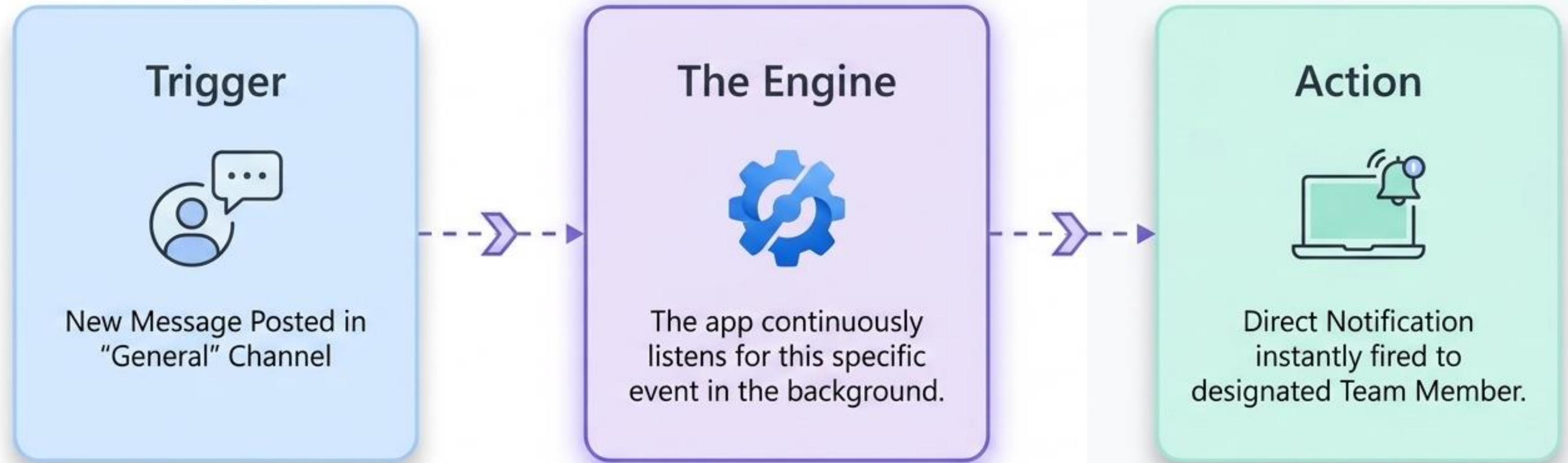
Click the Workflows icon to open the new v2 dashboard (featuring Home, Templates, and Chat tabs).



The Path to Your First Automation



Blueprint: Channel Notification Automation



Live Demonstration: Watch the Screen



The Demo Scenario

Trainer will build: "Help me prepare for my day" template.

Result: An automated 8:00 AM daily briefing of meetings and tasks sent via Teams chat.

What to Observe

- ✓ Browsing the "Templates" vs "Build from scratch" tabs.
- ✓ Setting the correct Time Zone (crucial for scheduled triggers).
- ✓ Renaming the workflow before saving so it is easily identifiable.
- ✓ Approving app connection permissions ("Sign in").

Hands-On Exercise: Build Your Channel Notification

Your Mission:

When someone posts in a selected channel, send me a direct Teams notification.

1. Open Teams -> Apps -> Workflows.
2. Search the template library for: "Notify me when..."
3. Configure the drop-downs to match your specific testing channel.
4. Save the workflow.
5. Test it: Post a real message in that channel and wait for the ping!



**Estimated time:
5–10 minutes.**

Live Troubleshooting & Workflow Tips



Common Mistakes

- Forgetting to set the correct Time Zone in scheduled workflows (defaults to UTC).
- Selecting the wrong Team/Channel in the dropdown menus.
- Forgetting to rename the workflow from its default string.



Where to Click

Look for the blue “Add connection” or “Sign in” buttons.

Teams requires you to explicitly grant permission the very first time you use Power Automate.



Pro Tip

Always click the “Run History” tab inside your workflow details.

It will explicitly tell you if a trigger succeeded or failed, and exactly why.

5 Ways to Automate Your Workday

1 Automatic Meeting Reminder: "Please provide an update on your tasks before the 10 AM stand-up."



2 Approval Request Automation: Route leave or document approvals automatically to managers via Teams chat.



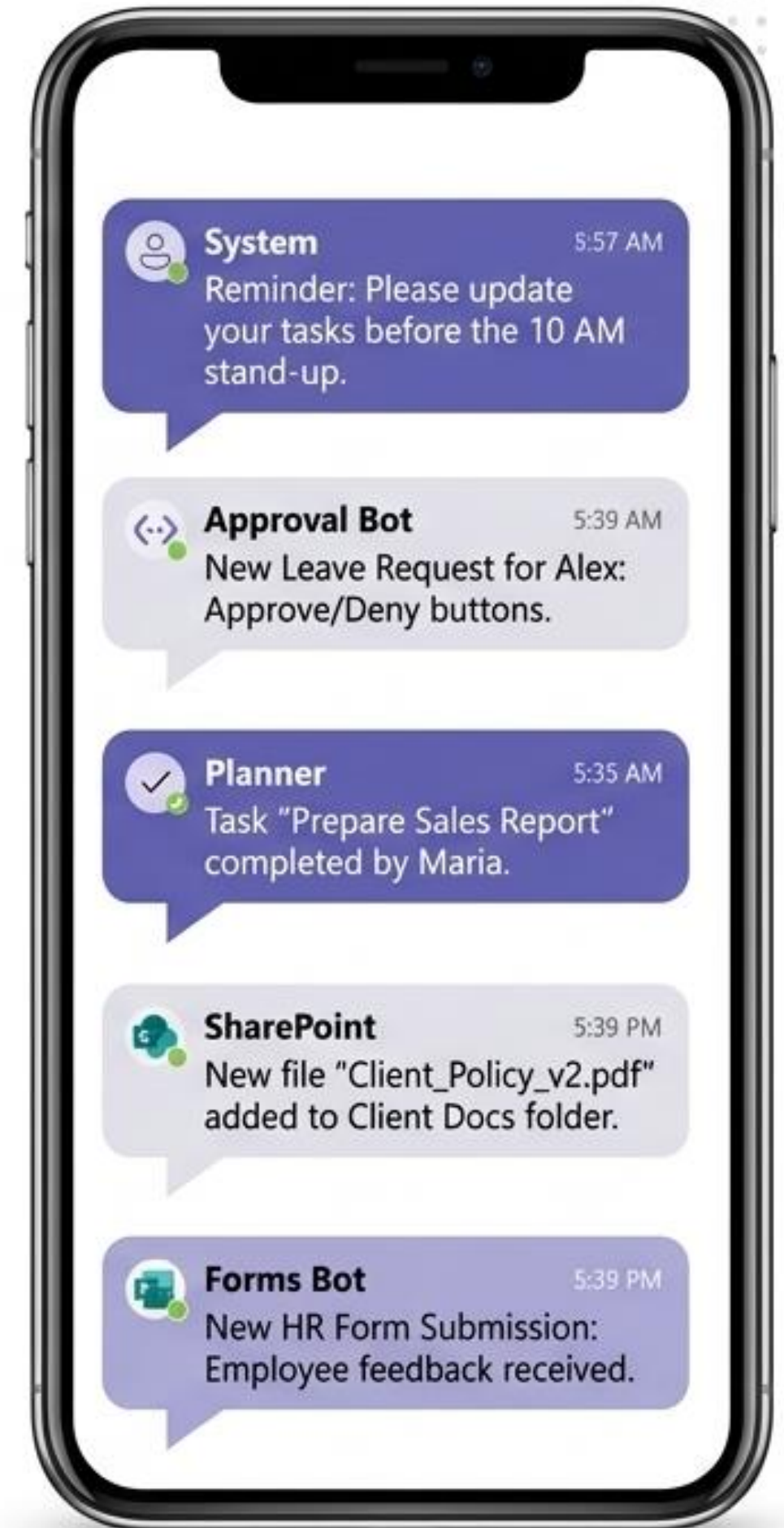
3 Task Assignment Visibility: Notify a channel the exact moment a Planner/To-Do task is marked complete.



4 File Upload Alert: Get pinged instantly when a client drops a new policy document into a SharePoint folder.



5 Form Submission Notification: Receive a direct chat summary the second an internal HR or IT support form is submitted.



Expanding the Ecosystem: Advanced Integrations



Microsoft Forms

Turn static survey responses into actionable Teams tasks or channel announcements the moment they are submitted.



SharePoint

Track specific document library changes. Never manually check if a file has been updated again.



Approvals

Build multi-step approval chains that allow managers to click 'Approve' or 'Reject' directly inside a Teams chat bubble.



AI & Copilot

For users with Copilot licenses, build complex Workflows simply by typing natural language prompts in the 'Chat' tab.

Choose Your Next Workflow Build



Option 1: Scheduled Meeting Reminder

Description: Schedule a recurring weekly message to prompt your team for updates.

Expected Outcome:
Never chase colleagues for routine status updates again.



Option 2: Planner Task Automation

Description: Link Microsoft Planner to Teams to announce completed work automatically.

Expected Outcome:
Hands-free project visibility and morale boosting.



Option 3: Form Notification

Description: Alert a specific channel with summarized data when a Form is filled out.

Expected Outcome:
Drastically faster response times to internal data requests.

Essential Rules for Teams Workflows



Workflows Live Inside Teams

There is no need to open web browsers or secondary software. The Home, Templates, and Chat tabs inside the Teams app handle everything.



Templates Are Your Best Friend

Never start from scratch unless necessary. The vast catalog of pre-built triggers covers 90% of business use cases.



Think "Trigger -> Action"

Train your brain to recognize repetitive friction. If a process starts with "When X happens..." it can be automated.



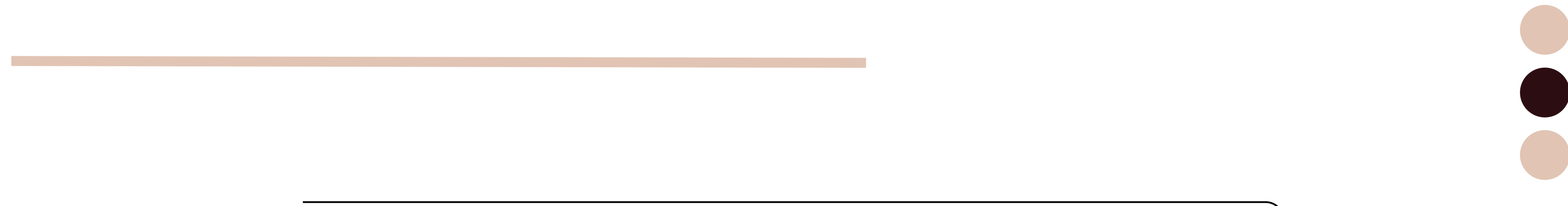
The Compound Effect of the AI Mindset



The diagram illustrates the compound effect of an AI mindset. It features a gear icon representing a task, a large 'X' for multiplication, a calendar icon representing workdays, an equals sign, and a large upward-pointing arrow representing reclaimed time. The text '5 Minutes Saved Daily' is positioned below the gear, '250 Workdays' below the calendar, and '20 Hours Reclaimed Per Year' to the right of the equals sign.

$$\begin{array}{ccccc} \text{5 Minutes} & \times & \text{250} & = & \text{20 Hours} \\ \text{Saved Daily} & & \text{Workdays} & & \text{Reclaimed Per Year} \end{array}$$

You cannot break Microsoft Teams by experimenting with Workflows.
Look at your calendar tomorrow. Find one recurring task. Automate it.
Welcome to the future of work at Prudential Takaful.



QUESTIONS & ANSWERS



www.reallygreatsite.com



THANK YOU



By Olivia Wilson

